This report describes the results of automatic processing of a large number of scientific papers according to a rigorously defined criterion of coupling. The population of papers under study was ordered into groups that satisfy the stated criterion of interrelation. An examination of the papers that constitute the groups shows a high degree of logical correlation.

Bibliographic Coupling Between Scientific Papers[1]

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A previous report described a new method for grouping technical and scientific papers on the basis of bibliographic coupling units. A single item of reference used by two papers was defined as a unit of coupling between them. Based on this unit, two graded criteria of coupling were defined.[2]

Criterion A: A number of papers constitute a related group G_A if each member of the group has at least one coupling unit to a given test paper P_o. The coupling strength between P_o and any member of G_A is measured by the number of coupling units (n) between them. G_A^n is that portion of G_A that is linked to P_o through n coupling units.

Criterion B: A number of papers constitute a related group G_B if each member of the group has at least one coupling unit to every other member of the group.

A computer program was written to process a large number of scientific articles into groups that satisfy Criterion A.[3] The population of papers under study consisted of 36 volumes of the *Physical Review* (Vols. 77 to 112 inclusive). These volumes contained 8,521 articles which generated 137,000 references to 795 separate sources.[4] From each article in this range of the *Physical Review* the following information was extracted and put on magnetic tape:

1. Volume and page number on which the article begins
2. Title
3. Author(s)
4. All bibliographic data in the form
   a. Title of journal or other source (coded by number)
   b. Volume number (if periodical)
   c. Page number

This data completely identifies each bibliographic item except for the rare case of two papers starting on the same page. When that happens, the computer will interpret the two papers as identical and thus introduce false couplings into the group. Although a few errors due to this anomaly were found in our results, it is a rare phenomenon that will be corrected in future work.

Let us call P_m the population of papers being examined and P_o the test paper. Several strategies are available for the ordering of members of P_m according to Criterion A.

1. One or more papers outside of P_m may serve as P_o's, and all or part of P_m may be processed. Example: Search *Physical Review,* 1958, for papers that match a given paper.

2. Part of P_m may constitute P_o's and all or part of the remaining P_m may be searched. Example: Search the past five years of the *Physical Review* for papers that match certain papers in the latest number.

3. All or part of P_m may be arranged into related groups, each item of P_m serving in turn as P_o for all the others. Example: Arrange a given volume of *Physical Review* into related groups.

The results of such processing applied to our entire set of P_m, 8,521 papers, are clearly too voluminous to be presented in a report. We give here the detailed results of strategy 3 applied to Volume 97 of the *Physical Review.* The volume contains 265 articles (see Table 1). Each article in this volume was treated as P_o and the remaining 264 as P_m. Thus 265 G_A's were generated. A

[1] Supported by a grant from the National Science Foundation to the Massachusetts Institute of Technology Libraries.

[2] The possibility of grouping scientific papers on the basis of *use* rather than *content* was suggested by Professor R. M. Fano in *Documentation in Action,* 1956, chapter XIV-e, pp. 238–244, Reinhold Publishing Corp., New York; and independently by the present author in Concerning some problems of intrascience communication, 1958, *Lincoln Laboratory Group Report 45–35,* 8 December.

[3] I wish to acknowledge the assistance of Mr. David Friedman and Mr. Armen Gabrielian who wrote the program for the IBM 709 computer.

[4] Preliminary statistics on the source material were published in M. M. Kessler, Technical information flow patterns, 1961. *Proc. Western Joint Computer Conference,* pp. 247–257. A detailed analysis is being prepared.

sample of the actual computer output is shown in Table 2. The print-out lists each article in Vol. 97 as P_o (first column). Then follows a list of all other papers in the volume that are related to it through one or more bibliographic couplings (column three). The articles that constitute the coupling units are listed in column two of the print-out.

The number of articles in each G_A varied from zero (not coupled to any other paper in Vol. 97) to 27. The number of coupling units observed varied from one, the weakest coupling, to eleven. The number of articles in the various G_A's and the frequency of coupling bonds are shown in Table 3 and Table 4. Table 5 shows all the G_A's generated from the papers in Vol. 97. Table 6 gives the titles and authors of the fourteen most strongly coupled articles. Table 7 gives the titles and authors of all the papers in every tenth G_A taken from Table 5.

It is suggested that after reading the text the reader may consult Table 6 for dramatic examples of bibliographic coupling. He may then proceed to Table 7 for a representative sample of G_A's. Using Table 5 in conjunction with the table of contents (Table 1), the reader may examine all the G_A's generated from Vol. 97.

The results of this experiment prove the existence of the G_A phenomenon in a population of well-edited papers in a well-established field of science. Extension of this processing scheme beyond *Physical Review* articles is under study.

Although the power of the method and its utility to an actual working system are yet to be assessed, several remarks may be made concerning its properties as a retrieval tool.

1. It is independent of words and language. All the processing is done in terms of numbers. We thus avoid all the difficulties of language, syntax and word habits.
2. No expert reading or judgment is required. Indeed, the text need not even be available.
3. The group of papers associated with a given test paper extend into the past as well as the future.
4. The method does not produce a static classification or permanent index number for a given paper. The groupings will undergo changes that reflect the current usages and interests of the scientific community.
5. The member papers of a G_A may be considered as the "logical references" of P_o. A population of papers, P_m, may thus be produced in such a way that for each paper in P_m we substitute the members of its G_A for the actual references. (That is, replace the actual references of the paper by its logical references.) This population of papers may then be further processed according to Criterion A. The significance of this logic is being investigated.

TABLE 1. Table of Contents of *Physical Review* Vol. 97, 1955

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TABLE 2. Sample of Computer Print-Out

ARTICLE 0010971538
0010971538 SHARES SOURCE 1030120314 WITH ARTICLE 0010970322
0010971538 SHARES SOURCE 1030120362 WITH ARTICLE 0010970322
0010971538 SHARES SOURCE 0010920958 WITH ARTICLE 0010970322
0010971538 SHARES SOURCE 1030120362 WITH ARTICLE 0010970587
0010971538 SHARES SOURCE 0010971538 WITH ARTICLE 0010971525
0010971538 SHARES SOURCE 1030120362 WITH ARTICLE 0010971525
0010971538 SHARES SOURCE 0010929857 WITH ARTICLE 0010971525
0010971538 SHARES SOURCE 0010971531 WITH ARTICLE 0010971525
0010971538 SHARES SOURCE 0120230015 WITH ARTICLE 0010971525
0010971538 SHARES SOURCE 0010971538 WITH ARTICLE 0010971531
0010971538 SHARES SOURCE 1030120314 WITH ARTICLE 0010971531
0010971538 SHARES SOURCE 1030120362 WITH ARTICLE 0010971531
0010971538 SHARES SOURCE 0010920857 WITH ARTICLE 0010971531
0010971538 SHARES SOURCE 0010971525 WITH ARTICLE 0010971531
0010971538 SHARES SOURCE 1030120362 WITH ARTICLE 0010971596

ARTICLE 0010971544

ARTICLE 0010971557
0010971557 SHARES SOURCE 0010900644 WITH ARTICLE 0010970733
0010971557 SHARES SOURCE 0010911571 WITH ARTICLE 0010970733
0010971557 SHARES SOURCE 0030629806 WITH ARTICLE 0010970916
0010971557 SHARES SOURCE 0010920609 WITH ARTICLE 0010971509

ARTICLE 0010971558
0010971558 SHARES SOURCE 050080153 WITH ARTICLE 00109700055
0010971558 SHARES SOURCE 0010941411 WITH ARTICLE 0010970060
0010971558 SHARES SOURCE 0500090153 WITH ARTICLE 0010970060
0010971558 SHARES SOURCE 0010940544 WITH ARTICLE 0010970060

ARTICLE 0010971567
0010971567 SHARES SOURCE 123150552 WITH ARTICLE 0010970629
0010971567 SHARES SOURCE 0231540207 WITH ARTICLE 0010970629

ARTICLE 0010971575
0010971575 SHARES SOURCE 1010060762 WITH ARTICLE 0010970607

ARTICLE 0010971579

ARTICLE 0010971584
0010971584 SHARES SOURCE 0010910812 WITH ARTICLE 0010970079

ARTICLE 0010971591
0010971591 SHARES SOURCE 0300260007 WITH ARTICLE 0010970277
0010971591 SHARES SOURCE 0300260007 WITH ARTICLE 0010970347
0010971591 SHARES SOURCE 0940010043 WITH ARTICLE 0010970355
0010971591 SHARES SOURCE 1300260007 WITH ARTICLE 0010970355
0010971591 SHARES SOURCE 0300180384 WITH ARTICLE 0010970676

TABLE 3. Number of Groups Containing N Articles

Number of Articles (N)	Number of GA's Containing N Articles
0	50
1	25
2	34
3	35
4	22
5	15
6	16
7	13
8	8
9	2
10	7
11	8
12	3
13	5
14	3
15	4
16	1
17	1
18	1
19	1
20	2
22	2
25	1
26	2
27	1

TABLE 4. List of Coupled Papers of Strength Two and Higher *

n = 11		n = 9		n = 8	
109	991	188	1283	142	689
419	708	1059	1077		
		1077	1314		
n = 7		**n = 6**			
89	1191	55	60		
290	363	355	1620		
869	1647	1474	1490		
1059	1314	1474	1509		
n = 5		**n = 5**		**n = 5**	
151	166	492	1162	1051	1059
159	206	492	1314	1051	1077
181	797	542	1191	1059	1336
224	985	736	991	1212	1267
411	937	891	1620	1429	1447
n = 4		**n = 4**		**n = 4**	
84	1661	224	1366	1110	1327
93	1255	322	1531	1191	1226
109	736	434	1327	1279	1283
122	492	769	1292	1525	1531
151	159	937	951	1525	1538
151	206	1007	1191	1531	1538
166	172	1031	1212		

TABLE 4 — Continued

n = 3		n = 3		n = 3	
1	9	322	1538	1077	1305
60	1558	355	891	1077	1336
93	117	542	1199	1092	1327
93	1272	614	669	1131	1139
103	1191	629	1419	1162	1314
104	1226	680	684	1205	1266
109	463	783	1077	1255	1272
117	1272	783	1305	1261	1679
122	1327	797	1086	1344	1366
172	193	869	916	1353	1366
188	1279	937	1654	1490	1509
224	1199	1007	1023	542	1226
322	1525	1017	1025		
n = 2		**n = 2**		**n = 2**	
25	29	181	1086	977	1017
76	1212	193	206	977	1255
89	419	311	322	985	1205
89	542	355	1591	985	1222
89	1226	380	680	1004	1661
89	1255	411	951	1007	1272
89	1272	451	1255	1017	1255
93	97	463	991	1047	1051
93	451	463	1272	1047	1077
93	977	629	1567	1059	1344
93	1017	680	1654	1191	1199
97	1199	680	1661	1199	1226
97	1272	684	1654	1199	1366
97	1380	684	1661	1212	1255
104	1249	701	1199	1237	1245
109	267	701	1380	1272	1366
117	977	720	1226	1283	1305
117	1255	733	1557	1283	1314
122	970	780	1303	1327	1336
122	1314	783	1314	1336	1344
132	1199	791	797	1336	1353
137	766	823	1162	1336	1366
142	463	855	1429	1344	1353
142	1191	891	897	1344	1490
159	166	933	1474	1344	1509
224	1353	937	967	1399	1453
224	1222	937	1474	1429	1453
267	275	951	1654	1654	1661

* Each number refers to the title page of an article in Vol. 97 of *Physical Review*. Each pair of articles is related through the number of coupling units, n, indicated. There are 450 pairs of papers related through one coupling unit, n = 1). These are not listed.

TABLE 5. Groups of Papers in *Physical Review*, Vol. 97 (1955) Related Through Criterion A*

A1	9.4 22 25 855 902 1429 1447
A9	1.4
A13	
A22	1 855 1429 1453
A25	1 855 902 1429.2 1447 1611
A33	372 507 1399
A39	290 355 363 654 891 1469 1474
A52	166 916
A55	60.7 558
A60	55.7 558.3

TABLE 5 — Continued

A66	
A74	578 902
A76	212.2
A79	1584
A84	967 1004 1661.5
A86	
A88	
A89	93 97 103 117 121 404 419.2 427 428 451 463 542.2 708 757 991 1007 1071 1092 1191.7 1199 1226.2 1245 1255.2 1272.2 1380
A93	89 97.2 103 117.3 121 139 427 432 451.2 463 750 975 977.2 987 1017.2 1031 1092 1212 1245 1255.4 1272.3 1380
A97	89 93.2 109 117 121 142 224 419 427 451 463 542 701 1031 1092 1191 1199.2 1226 1237 1255 1266 1267 1272.2 1353 1366 1380.2
A103	89 93 542 1191.3 1226 1255 1272 1380
A104	224 720 744 1017 1191 1205 1226.3 1237 1245 1249.2
A109	97 267.2 463.3 736.4 766 991.11
A117	89 93.3 97 121 427 451
A121	89 93 97 117 427 451 463 1092 1191 1255 1272
A122	444 492.4 508 970.3 1162 1314.2 1327.3
A131	132 224
A132	131 224 1199.2 1249
A137	511 766.2
A139	93 1025
A142	97 224 463.2 542 689.8 701 937 1007 1191.2 1199 1226 1272 1353 1366 1380
A151	159.4 166.5 206.4
A159	151.4 166.2 172 206.5 1084
A166	52 151.5 159.3 172.4 780 1084 1110 1303
A172	159 166.4 193.3 206 1084
A181	736 797.5 1086.3 1387
A184	975 1205 1314
A188	434 469 970 1279.3 1283.10 1305 1327
A193	172.3 206.2 783
A201	
A206	151.4 159.5 172 193.2
A212	
A224	97 104 131 132 142 456 542 701 985.5 1002 1051 1059 1077 1110 1191 1199.3 1205 1222.2 1226 1237 1245 1336 1344 1353.2 1366.4 1380
A255	
A259	263.2
A263	259 855 869 1647
A267	109.2 275.3
A275	267.3
A277	347 355 1591
A288	
A290	39 363.7
A304	
A311	322.2 354 469 883 469
A322	311.2 587 1525.3 1531.4 1538.3 1596
A334	
A345	
A347	277 355 1591
A352	869 916 1647

TABLE 5 — Continued

A354	311 883
A355	39 277 347 486 660 676 981.3 897 1591.2 1620.6
A363	39 290.7 654 916 933 1474 1490
A372	33 1162
A380	463 680.2 686 883
A395	486 937 951 1654
A404	89 428 1071 1191 1237 1636
A411	937.5 951.2 967 474 1671
A419	89.2 97 708.12 986
A427	89 93 97 117 121 451 463 1092 1255 1272
A428	89 404 1004 1071 1191 1272 1380
A432	93
A434	188 444 1327.4
A444	122 434 736
A446	701 1199
A451	89 93.2 97 117 121 427 463 977 987 1017 1031 1092 1212 1255.2 1272
A456	224
A463	89 93 97 109.3 117 121 142.2 380 427 451 686 736 991.2 1007 1031 1092 1191 1222 1255 1272.2
A469	188 311 809 970 1327
A470	
A474	
A480	791 902 1059 1071 1336 1344
A486	355 395 791 1071
A492	122.4 1162.5 1314.5
A505	508 1158
A507	33 1399
A508	122 505 970
A511	137 766 1139 1146
A536	
A540	
A542	89.2 97 103 142 224 701 1059 1191.5 1199.3 1226.3 1353 1366 1380
A571	
A578	74
A583	
A587	322 869 891 916 1525 1531 1538 1596
A599	1611 1636
A607	1575
A614	669.3 1469
A629	1419.4 1567.2
A641	1525
A647	869 902
A654	39 363
A660	355 891 1390 1620
A666	
A669	614.3 1469
A673	
A676	355 1521 1591
A680	380.2 684.3 1654.2 1661.2
A684	680.3 1654.2 1661.2
A686	380 463 1199
A689	142.8 916 937
A701	97 142 224 446 542 1191 1199.2 1226 1353 1366 1380.2
A707	1092
A708	89 419.11 986
A715	117 977 1031 1272

TABLE 5 — Continued

A720	104 744 1025 1191 1205 1226.2 1237 1245 1249
A726	
A733	1557.2
A736	109.4 181 444 463 991.5 1071
A743	1327
A744	104 720 1191 1205 1226 1237 1245 1249
A750	93 1222
A753	
A757	89 1245
A766	109 137.2 511
A769	783 1292.4
A780	166 1303.3 1327
A783	193 769 1059 1077.3 1305.3 1314.2
A791	480 486 797.2
A797	181.5 791.2 1059 1086.3 1110 1283 1292 1314 1387
A809	469
A810	
A823	1162.2 1390
A841	
A849	
A855	1 22 25 263 869 902 1429.2 1447 1453 1647
A863	
A869	263 352 587 647 855 883 891 916.3 1474 1647.7
A883	311 354 380 1031 1647
A889	
A891	39 355.3 587 660 869 897.2 916 937 1469 1620.5
A897	355 891.2 916 1474
A902	1 25 74 80 647 855 1429 1447
A916	52 352 363 587 689 869.3 891 897 933 1474 1557 1654
A926	
A933	363 916 1474.2 1490
A937	142 395 411.5 689 891 951.5 967.2 1474.2 1509 1654.3
A951	395 411.2 937.4 967 1654.2
A967	84 411 937.2 951 1661 1671
A970	122.2 188.1 469 508
A975	93 117 184 977 1017 1205 1255
A977	93.2 117.2 451 715 975 987 1017.2 1031 1212 1255.2 1272
A985	224.5 119 1205.2 1222.2 1366
A986	419 708
A987	93 451 977 1017 1031 1212 1255
A991	89 109.11 463.2 736.5
A1002	224 1249
A1004	84 428 1226 1234 1249 1661.2
A1007	89 142 463 1023.3 1191.4 1255 1272.2
A1017	93.2 104 117 451 975 977.2 987 1025.3 1031 1212 1226 1255.2
A1023	1007.3 1191
A1025	139 720 1017.3
A1031	93 97 451 463 715 883 977 987 1017 1212.4 1255 1267 1272
A1033	1327
A1037	1679
A1038	
A1043	

TABLE 5 — Continued

A1047	1051.2 1077.2 1145
A1051	224 1047 1059.5 1077.5 1336
A1059	224 480 542 783 797 1051.5 1077.10 1092 1199 1283 1305 1314.7 1327 1336.5 1344.2 1353 1366
A1071	89 404 428 480 486 736 1191
A1077	224 783.3 1047.2 1051.5 1059.10 1305.3 1314.9 1336.3
A1084	159 166 172 1162
A1086	181.2 797.4
A1092	89 93 97 117 121 427 451 463 707 1059 1110 1255 1272 1327.3 1336
A1110	166 224 797 1092 1292 1327.4
A1123	
A1126	
A1131	1139.4 1406
A1139	511 1131.3 1146
A1144	
A1145	1047
A1146	511 1139
A1158	505 1712
A1162	122 372 492.5 823.2 1084 1314.3 1390
A1191	89.7 97 103.3 104 121 142.2 224 404 428 463 542.5 701 720 744 1007.4 1023 1071 1199.2 1205 1212 1226.4 1237 1245 1249 1272 1353 1366 1380
A1199	89 97.2 132.2 142 224.3 446 542.3 686 701.2 985 1059 1191.2 1222 1226.2 1249 1353 1366.2 1380
A1205	104 184 224 720 744 975 985.2 1191 1226 1237 1245 1249 1266.3
A1212	76.2 93 451 977 987 1017 1031.4 1191 1255.2 1267.5 1272
A1222	224.2 463 750 985.2 1199 1366
A1226	89.2 97 104.3 142 224 542.3 701 720.2 744 1004 1017 1191.4 1199.2 1205 1237 1245 1249 1353 1366 1380
A1234	1004
A1237	97 104 224 404 720 744 1191 1205 1226 1245.2 1249
A1245	89 93 104 224 720 744 757 1191 1205 1226 1237.2 1249 1255
A1249	104.2 132 720 744 1002 1004 1191 1199 1205 1226 1237 1245
A1255	89.2 93.4 97 103 117.2 121 427 451.2 463 975 977.2 987 1007 1017.2 1031 1092 1212.2 1245 1272.3
A1258	
A1261	1446 1679.3 1699
A1266	97 1205.4
A1267	97 1031 1212.5 1272
A1272	89.2 93.3 97.2 103 117.3 121 142 427 428 451 463.2 715 977 1007.2 1031 1092 1191 1212 1255.3 1267 1344 1366.2 1380
A1276	
A1279	188.3 1283.5 1305
A1283	188.9 797 1059 1279.5 1305.2 1314.2 1327
A1292	769.4 797 1110
A1303	166 780.2 1327
A1305	188 783.3 1059 1077.3 1279 1283.2 1314
A1314	122.2 184 492.5 783.2 797 1059.7 1077.9 1162.3 1283.2 1305 1327 1336

TABLE 5 — Continued

A1325	
A1327	122.3 188 433.4 469 743 780 1033 1059 1092.3 1110.4 1283 1303 1314 1336.2
A1336	224 480 1051 1059.5 1077.3 1092 1314 1327.2 1344.2 1353.2 1366.2
A1344	224 480 1059.2 1272 1336.2 1353.2 1366.4 1474 1490.2 1509.2 1653
A1353	97 142 224.2 542 701 1059 1191 1199 1226 1336.2 1344.3 1366.3 1380
A1366	97 142 224.4 542 701 985 1059 1191 1199.2 1222 1226 1272.2 1336.2 1344.4 1353.3 1380
A1380	89 93 97.2 103 142 224 428 542 701.2 1191 1199 1226 1272 1353 1366
A1387	181 797 1086
A1390	660 823 1162 1399
A1392	
A1399	33 507 1390 1429 1453.2
A1406	1131
A1413	
A1419	629.3
A1421	
A1429	1 22 25.2 855.2 902 1399 1447.5 1453.2
A1435	
A1443	
A1446	1261
A1447	1 25 855 902 1429.5
A1453	22 855 1399.2 1429.2
A1456	
A1463	
A1469	39 311 614 669 891
A1471	
A1474	39 363 411 869 897 916 933.2 937.2 1344 1490.6 1509.7
A1490	363 933 1344.2 1474.6 1509.4 1653
A1509	937 1344.2 1474.6 1490.3 1557 1653
A1521	676 1647
A1525	322.3 587 641 1531.5 1538.5 1596
A1531	322.4 587 1525.5 1538.5 1596

TABLE 5 — Continued

A1538	322.3 587 1525.5 1531.5 1596
A1544	
A1557	733.2 916 1509
A1558	55 60.3
A1567	629.2
A1575	607
A1579	
A1584	79
A1591	277 347 355.2 676
A1596	322 587 1525 1531 1538
A1611	25 599 1636
A1620	355.6 660 891.5
A1627	
A1634	
A1636	404 599 1611
A1647	263 352 855 869.7 883 1521
A1653	1344 1490 1509
A1654	395 680.2 684.2 916 937.3 951.2 1661.2 1671
A1661	84.7 680.2 684.2 967 1004.2 1654.2
A1664	
A1668	
A1671	411 967 1654
A1679	1037 1261.3 1699
A1699	1261 1679
A1710	
A1712	1158

* Each number in the table refers to the title page of an article in *Physical Review*, Vol. 97, 1955. A159 indicates that the article starting on page 159 is the test paper, P_o, for the subsequent group of numbers. The numbers that follow each test article refer to papers that are related to it through Criterion A. The number that follows a decimal point indicates the coupling strength, n, between this article and P_o. If no decimal point is indicated, it is to be understood that $n = 1$.

Example:

A159 151.4 166.2 172 206.5 1084

The article in Vol. 97, p. 159, was used as the test paper, P_o. The program generated a G_A consisting of 5 papers. The paper that starts on page 151 in Vol. 97 was coupled to P_o through 4 coupling units; paper on page 166, two coupling units; page 172, one coupling unit; p. 206, 5 units; p. 1084, one unit.

TABLE 6. Pairs of Papers Most Strongly Coupled *

109.11.991

Electron-Neutrino Angular Correlation in the Beta Decay of Neon 19.....D. R. Maxson, J. S. Allen, and W. K. Jentschke

Gamow-Teller Interaction in the Decay of He^6.....B. M. Rustad and S. L. Ruby

419.11.708

Internal and External Bremsstrahlung Accompanying the Beta Rays of P^{32}.....K. Liden and N. Starfelt

Internal and External Bremsstrahlung in Connection with the Beta Decay of S^{35}.....N. Starfelt and N. L. Svantesson

188.9.1283

Photoproduction of π^0 Mesons from Protons.....Y. Goldschmidt-Clermont, L. S. Osborne, and M. Scott

Photoproduction of Neutral Pions in Hydrogen: Magnetic Analysis of Recoil Protons.....D. C. Oakley and R. L. Walker

1059.9.1077

Theory of Polarization of Nucleons Scattered Elastically by Nuclei.....Sidney Fernbach, Warren Heckrotte and Joseph Lepore

Polarization in Scattering by Complex Nuclei.....S. Tamor

1077.9.1314

Polarization in Scattering by Complex Nuclei.....S. Tamor

Polarization of Nucleons Elastically Scattered from Nuclei.....R. M. Sternheimer

TABLE 6 — Continued

142.8.689
Gamma and X-Radiation in the Decay of Am^{241}.....H. Jaffe, T. O. Passell, C. I. Browne, and I. Perlman
Electromagnetic Spectrum of Am^{241}.....Paul P. Day

89.7.1191
Coulomb Excitation of Neodymium.....B. E. Simmons, D. M. Van Patter, K. F. Famularo, and R. V. Stuart
Electric Excitation of Heavy Nuclei by Protons.....Clyde McClelland, Hans Mark, and Clark Goodman

290.7.363
Dynamics of Simple Lattices.....Herbert B. Rosenstock
Vibration Spectra and Specific Heats of Cubic Metals. I. Theory and Application to Sodium....A. B. Bhatia

869.7.1647
Motions of Electrons and Holes in Perturbed Fields.....J. M. Luttinger and W. Kohn
Theory of the Infrared Absorption of Carriers in Germanium and Silicon.....A. H. Kahn

1059.7.1314
Theory of Polarization of Nucleons Scattered Elastically by Nuclei.....
.....Sidney Fernbach, Warren Heckrotte and J. Lepore
Polarization of Nucleons Elastically Scattered from Nuclei.....R. M. Sternheimer

55.6.60
Domain Rotation in Nickel Ferrite.....Fielding Brown and Charles L. Gravel
Magnetic Rotation Phenomena in a Polycrystalline Ferrite.....David Park

355.6.1620
Temperature Dependence of Electron Mobility in AgCl.....Frederick C. Brown
Mobility of Electrons and Holes in the Polar Crystal, PbS.....Richard L. Petritz and Wayne W. Scanlon

1474.6.1490
Quantum Theory of Many-Particle Systems. I. Physical Interpretations by Means of Density Matrices, Natural Spin-Orbitals, and Convergence Problems in the Method of Configurational Interaction.....Per-Olov Lowdin
Quantum Theory of Many-Particle Systems. II. Study of the Ordinary Hartree-Fock Approximation.....
.....Per-Olov Lowdin

1474.6.1509
Quantum Theory of Many-Particle Systems. I. Physical Interpretations by Means of Density Matrices, Natural Spin-Orbitals, and Convergence Problems in the Method of Configurational Interaction..Per-Olov Lowdin
Quantum Theory of Many-Particle Systems. III. Extension of the Hartree-Fock Scheme to Include Degenerate Systems and Correlation Effects.....Per-Olov Lowdin

• Explanation: 109.11.991 means: paper that begins on page 109 is coupled through eleven units to paper that begins on page 991. The titles of the two papers follow each set of numbers.

TABLE 7. Detailed Description of Every Tenth Group in TABLE 5 (arranged in order of decreasing n)

$P_o = A1$ Pressure-Volume Isotherms of He^4 below 4.2°K.....William E. Keller
9.4 Second Virial Coefficients of Helium from the Exp-Six Potential.....
.....John E. Kilpatrick, William E. Keller, and Edward F. Hammel
22.1 Thickness of the Helium Film as a Function of Height.....Lothar Meyer
25.1 Flow of Helium through Narrow Slits.....R. T. Swim and H. E. Rorschach, Jr.
855.1 Transmission of Slow Neutrons by Liquid Helium.....
.....Henry S. Sommers, Jr., J. Gregory Dash, and Louis Goldstein
902.1 Thermal Conductivity of Indium-Thallium Alloys at Low Temperatures.....Ronald J. Sladek
1429.1 Coefficient of Expansion of Liquid Helium II.....K. R. Atkins and M. H. Edwards
1447.1 Ultrasonic Propagation in Magnetically Cooled Helium.....C. E. Chase and Melvin A. Herlin

$P_o = A66$ Effect of Strong Electrostatic Fields on the Resistance of Tungsten Wires in High Vacua.....
.....W. J. Deshotels and A. H. Weber

$P_o = A103$ Energy Levels of Hf^{177}.....P. Marmier and F. Boehm
1191.3 Electric Excitation of Heavy Nuclei by Protons.....Clyde McClelland, Hans Mark, and Clark Goodman
89.1 Coulomb Excitation of Neodymium.....B. E. Simmons, D. M. Van Patter, K. F. Famularo, and R. V. Stuart
93.1 Radioactivity of In^{117} and Sb^{117}.....Carl L. McGinnis
542.1 Coulomb Functions for Large Charges and Small Velocities.....
.....L. C. Biedenharn, R. L. Gluckstern, M. H. Hull, Jr., and G. Breit
1226.1 Gamma Radiation from the Inelastic Scattering of Protons by F^{19}.....C. A. Barnes
1255.1 Isomeric State of Y^{88} in the Decay of Zr^{88}.....E. K. Hyde, M. G. Florence, and A. E. Larsh
1272.1 Decay of Ti^{51} and Cr^{51}.....M. E. Bunker and J. W. Starner
1380.1 Magnetic Moments of Conjugate Nuclei.....J. G. Brennan and A. B. Volkov

TABLE 7 — Continued

$P_o = A142$	Gamma and X-Radiation in the Decay of Am^{241}	H. Jaffe, T. O. Passell, C. I. Browne, and I. Perlman
689.8	Electromagnetic Spectrum of Am^{241}	P. P. Day
463.2	β Spectrum of Bi^{210} (RaE) and the Coupling Constants of Scalar and Tensor Interactions in β Decay	G. E. Lee-Whiting
1191.2	Electric Excitation of Heavy Nuclei by Protons	Clyde McClelland, Hans Mark, and Clark Goodman
97.1	Decay Scheme of the Mirror Nucleus P^{29} and Related Results	H. Roderick, O. Lonsjo, and W. E. Meyerhof
224.1	Collision Matrices for the Compound Nucleus	R. G. Thomas
542.1	Coulomb Functions for Large Charges and Small Velocities	L. C. Biedenharn, R. L. Gluckstern, M. H. Hull, Jr., and G. Breit
701.1	Dynamics of Nuclear Deformation	D. R. Inglis
937.1	The Fine Structure of the Molecular Oxygen Ground State	M. Tinkham and M. W. P. Strandberg
1007.1	Decays of Ta^{182} and Ta^{183}	J. J. Murray, F. Boehm, P. Marmier, and J. W. M. DuMond
1199.1	Concept of Parentage of Nuclear States and Its Importance in Nuclear Reaction Phenomena	A. M. Lane and D. H. Wilkinson
1226.1	Gamma Radiation from the Inelastic Scattering of Protons by F^{19}	C. A. Barnes
1272.1	Decay of Ti^{51} and Cr^{51}	M. E. Bunker and J. W. Starner
1353.1	Two-Body Forces and Nuclear Saturation. III. Details of the Structure of the Nucleus	K. A. Brueckner
1366.1	Theory of Nuclear Models	R. J. Eden and N. C. Francis
1380.1	Magnetic Moments of Conjugate Nuclei	J. G. Brennan and A. B. Volkov
$P_o = A201$	Direct Detection of Soft Radiation above 50 Kilometers in the Auroral Zone	L. H. Meredith, M. B. Gottlieb, J. A. Van Allen
$P_o = A288$	Structure in Optical Absorption of Barium Oxide Films	R. J. Zollweg
$P_o = A355$	Temperature Dependence of Electron Mobility in AgCl	Frederick C. Brown
1620.6	Mobility of Electrons and Holes in the Polar Crystal, PbS	Richard L. Petritz and Wayne W. Scanlon
891.3	Thermal Ionization of Impurities in Polar Crystals. II. Application to Interstitials in Cubic ZnS	H. D. Vasileff
1591.2	Generation of Electron Traps by Plastic Flow in Alkali Halides	Masayasu Ueta and Werner Kanzig
39.1	Infrared Lattice Absorption in Ionic and Homopolar Crystals	Melvin Lax and Elias Burstein
277.1	F Centers in Pure and Hydride-Containing Alkali Halide Crystals	R. S. Alger and R. D. Jordan
347.1	Photostimulated Emission of Some Activated Alkali Halide Phosphors	C. E. Mandeville and H. O. Albrecht
486.1	Landau Distribution and Density Effect at High Gas Pressures	E. D. Palmatier, J. T. Meers, and C. M. Askey
660.1	Slow Electrons in a Polar Crystal	R. P. Feynman
676.1	Physical Defects in Silver Halides	Chester R. Berry
897.1	Effect of Overlap on Electrostatic Lattice Potentials in Ionic Crystals	Joseph Birman
$P_o = A432$	Beta Spectrum of C^{14}	A. V. Pohm, R. C. Waddell, J. P. Powers, and E. N. Jensen
93.1	Radioactivity of In^{117} and Sb^{117}	Carl L. McGinnis
$P_o = A480$	Scattering of 119-Mev Pions by Deuterium	D. E. Nagle
791.1	Scattering of 151-Mev Positive Pions by Protons	Raymond A. Grandey and Arnold F. Clark
902.1	Thermal Conductivity of Indium-Thallium Alloys at Low Temperatures	Ronald J. Sladek
1059.1	Theory of Polarization of Nucleons Scattered Elastically by Nuclei	Sidney Fernbach, Warren Heckrotte, and Joseph Lepore
1071.1	Specific Ionization by High-Energy Electrons	W. C. Barber
1336.1	Scattering of Fast Neutrons and Protons by Atomic Nuclei	Gyo Takeda and K. M. Watson
1344.1	Approximate Reduction of the Many-Body Problem for Strongly Interacting Particles to a Problem of Self-Consistent Fields	K. A. Brueckner and C. A. Levinson
$P_o = A571$	Thermodynamic Properties of He^3 and He^4 Solutions	V. S. Nanda
$P_o = A654$	Elastic Spectrum of Copper from Temperature-Diffuse Scattering of X-Rays	E. H. Jacobsen
39.1	Infrared Lattice Absorption in Ionic and Homopolar Crystals	Melvin Lax and Elias Burstein
363.1	Vibration Spectra and Specific Heats of Cubic Metals. I. Theory and Application to Sodium	A. B. Bhatia
$P_o = A701$	Dynamics of Nuclear Deformation	D. R. Inglis
1199.2	Concept of Parentage of Nuclear States and Its Importance in Nuclear Reaction Phenomena	A. M. Lane and D. H. Wilkinson
1380.2	Magnetic Moments of Conjugate Nuclei	J. G. Brennan and A. B. Volkov
97.1	Decay Scheme of the Mirror Nucleus P^{29} and Related Results	H. Roderick, O. Lonsjo, and W. E. Meyerhof
142.1	Gamma and X-Radiation in the Decay of Am^{241}	H. Jaffe, T. O. Passell, C. I. Browne, and I. Perlman

TABLE 7 — Continued

224.1 Collision Matrices for the Compound Nucleus....R. G. Thomas
446.1 Energy of the Ground State of Li^6....J. Irving and D. S. Schonland
542.1 Coulomb Functions for Large Charges and Small Velocities....L. C. Biedenharn, R. L. Gluckstern, M. H. Hull, Jr., and G. Breit
1191.1 Electric Excitation of Heavy Nuclei by Protons....Clyde McClelland, Hans Mark, and Clark Goodman
1226.1 Gamma Radiation from the Inelastic Scattering of Protons by F^{19}....C. A. Barnes
1353.1 Two-Body Forces and Nuclear Saturation. III. Details of the Structure of the Nucleus....K. A. Brueckner
1366.1 Theory of Nuclear Models....R. J. Eden and N. C. Francis

P_o = A750 Radioactivities of Zn^{69} and Zn^{71}....J. M. LeBlanc, J. M. Cork, and S. B. Burson
93.1 Radioactivity of In^{117} and Sb^{117}....Carl L. McGinnis
1222.1 Cross Section for Excitation of Pb^{207m} by Inelastic Scattering of Neutrons..P. H. Stelson and E. C. Campbell

P_o = A810 Biquadratic Spinor Identities....K. M. Case

P_o = A897 Effect of Overlap on Electrostatic Lattice Potentials in Ionic Crystals....Joseph Birman
891.2 Thermal Ionization of Impurities in Polar Crystals. II. Application to Interstitials in Cubic ZnS....H. D. Vasileff
355.1 Temperature Dependence of Electron Mobility in AgCl....Frederick C. Brown
916.1 X-Ray Spectroscopy of the Solid State: Potassium Chloride....L. G. Parratt and E. L. Jossem
1474.1 Quantum Theory of Many-Particle Systems. I. Physical Interpretations by Means of Density Matrices, Natural Spin-Orbitals, and Convergence Problems in the Method of Configurational Interaction....Per-Olov Lowdin

P_o = A977 Level Scheme of In^{115}....Jagdish Varma and C. E. Mandeville
93.2 Radioactivity of In^{117} and Sb^{117}....Carl L. McGinnis
117.2 Studies on the Isomeric Pair, Nb^{89m} and Nb^{89}....Hirdaya B. Mathur, Earl K. Hyde, Charles A. Levine, and Per K. Kofstad
1017.2 Gyromagnetic Ratio of an Excited State and Other Angular Correlation Measurements for Pb^{204m}....V. E. Krohn and S. Raboy
1255.1 Isomeric State of Y^{88} in the Decay of Zr^{88}....E. K. Hyde, M. G. Florence, and A. E. Larsh
451.1 Radioactive Decay of Cs^{134} and Cs^{134m}....G. L. Keister, E. B. Lee, and F. H. Schmidt
715.1 Energy Levels of Rh^{103} from the Decay of Pd^{103} and Ru^{103}....Babulal Saraf
975.1 Identification of Mo^{91} and Mo^{91m}....Peter Axel, John D. Fox, and Richard H. Parker
987.1 Disintegration of Te^{131}....Elizabeth Hebb
1031.1 Radiations from Cerium-141....James T. Jones, Jr., and E. N. Jensen
1212.1 Alignment of Cerium-141 and Neodymium-147 Nuclei....E. Ambler, R. P. Hudson, and G. M. Temmer
1272.1 Decay of Ti^{51} and Cr^{51}....M. E. Bunker and J. W. Starner

P_o = A1025 Geometrical Corrections in Angular Correlation Measurements....Arnold M. Feingold and Sherman Frankel
1017.3 Gyromagnetic Ratio of an Excited State and Other Angular Correlation Measurements for Pb^{204m}....V. E. Krohn and S. Raboy
139.1 Summing Studies of the Gamma Rays Following the Decays of Ba^{131}, Se^{75}, and Ru^{106}....D. C. Lu, W. H. Kelly, and M. L. Wiedenbeck
720.1 High-Energy Gamma Rays from the Proton Bombardment of Flourine....G. K. Farney, H. H. Givin, B. D. Kern, and T. M. Hahn

P_o = A1077 Polarization in Scattering by Complex Nuclei....S. Tamor
1059.10 Theory of Polarization of Nucleons Scattered Elastically by Nuclei....Sidney Fernbach, Warren Heckrotte, and Joseph Lepore
1314.9 Polarization of Nucleons Elastically Scattered from Nuclei....R. M. Sternheimer
1051.5 Polarization and Amplitudes in Nucleon-Nucleon Scattering....G. Breit, J. B. Ehrman, and M. H. Hull, Jr.
783.3 Proton-Proton Scattering at 437 Mev....R. B. Sutton, T. H. Fields, J. G. Fox, J. A. Kane, W. E. Mott, and R. A. Stallwood
1305.3 Reaction $p + p \rightarrow \pi^+ + d$....Frank S. Crawford, Jr. and M. L. Stevenson
1336.3 Scattering of Fast Neutrons and Protons by Atomic Nuclei....Gyo Takeda and K. M. Watson
1047.2 Amplitudes in Nucleon-Nucleon Scattering....G. Breit and M. H. Hull, Jr.
224.1 Collision Matrices for the Compound Nucleus....R. G. Thomas

P_o = A1145 Definition of the Virtual Level of the Deuteron....J. Humblet
1047.1 Amplitudes in Nucleon-Nucleon Scattering....G. Breit and M. H. Hull, Jr.

P_e = A1234 Search for Double Beta Decay in Ca^{48}....John A. McCarthy
1004.1 Masses of Light Nuclei....J. E. Drummond

TABLE 7 — Continued

P_o = A1276 High-Altitude Cosmic-Ray Neutron Density at the Geomagnetic Pole.....H. A. C. Neuburg, R. K. Soberman, M. J. Swetnick, and S. A. Korff

P_o = A1344 Approximate Reduction of the Many-Body Problem for Strongly Interacting Particles to a Problem of Self-Consistent Fields.....K. A. Brueckner and C. A. Levinson

1366.4 Theory of Nuclear Models.....R. J. Eden and N. C. Francis

1059.2 Theory of Polarization of Nucleons Scattered Elastically by Nuclei.....Sidney Fernbach, Warren Heckrotte, and Joseph V. Lepore

1336.2 Scattering of Fast Neutrons and Protons by Atomic Nuclei.....Gyo Takeda and K. M. Watson

1353.2 Two-Body Forces and Nuclear Saturation. III. Details of the Structure of the Nucleus.....K. A. Brueckner

1490.2 Quantum Theory of Many-Particle Systems. II. Study of the Ordinary Hartree-Fock Approximation.....Per-Olov Lowdin

1509.2 Quantum Theory of Many-Particle Systems. III. Extension of the Hartree-Fock Scheme to Include Degenerate Systems and Correlation Effects.....Per-Olov Lowdin

224.1 Collision Matrices for the Compound Nucleus.....R. G. Thomas

480.1 Scattering of 119-Mev Pions by Deuterium.....D. E. Nagle

1272.1 Decay of Ti^{51} and Cr^{51}.....M. E. Bunker and J. W. Starner

1474.1 Quantum Theory of Many-Particle Systems. I. Physical Interpretations by Means of Density Matrices, Natural Spin-Orbitals, and Convergence Problems in the Method of Configurational Interaction.....Per-Olov Lowdin

1653.1 New Treatment of the Auger Effect and the Fluorescence Yield in the Lighter Elements.....R. A. Rubenstein and J. N. Snyder

P_o = A1419 Relation Between Canonical and Microcanonical Ensembles.....Melvin Lax

629.3 Molecular Field in the Spherical Model.....Melvin Lax

P_o = A1469 Thermal Ionization and Capture of Electrons Trapped in Semiconductors..Hermann Gummel and Melvin Lax

39.1 Infrared Lattice Absorption in Ionic and Homopolar Crystals.....Melvin Lax and Elias Burstein

311.1 Trapping of Minority Carriers in Silicon. I. P-Type Silicon.....J. A. Hornbeck and J. R. Haynes

614.1 Gold as an Acceptor in Germanium.....W. C. Dunlap, Jr.

669.1 Properties of Germanium Doped with Cobalt.....W. W. Tyler, R. Newman, and H. H. Woodbury

891.1 Thermal Ionization of Impurities in Polar Crystals. II. Application to Interstitials in Cubic ZnS.....H. D. Vasileff

P_o = A1557 Thermalization of Positrons in Metals.....G. E. Lee-Whiting

733.2 Nature of Positron Annihilation in Liquids and Solids.....W. R. Dixon and L. E. H. Trainor

916.1 X-Ray Spectroscopy of the Solid State: Potassium Chloride.....L. G. Parratt and E. L. Jossem

1509.1 Quantum Theory of Many-Particle Systems. III. Extension of the Hartree-Fock Scheme to Include Degenerate Systems and Correlation Effects.....Per-Olov Lowdin

P_o = A1627 Paramagnetic Effect in Superconductors. I. Theoretical Aspects.....Hans Meissner

P_o = A1679 Nuclear Spin Exchange in Solids: Tl^{203} and Tl^{205} Magnetic Resonance in Thallium and Thallic Oxide.....N. Bloembergen and T. J. Rowland

1261.3 Effects of Perturbing Radiofrequency Fields on Nuclear Spin Coupling.....Arnold L. Bloom and James N. Shoolery

1037.1 Paramagnetic Resonance Absorption in Chromic Chloride....Yu Ting, L. D. Farringer, and Dudley Williams

1699.1 Free Magnetic Induction in Nuclear Quadrupole Resonance.....M. Bloom, E. L. Hahn, and B. Herzog

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